CS211 Final Exam Fall 2017 Name: 12/05/2017

1. What output is produced by the following code? (10points, no partial credits)

```
class A{
    int i = 1;
    A(){System.out.println(i);}
    public String x(){return "A.x";}
    public String y(){return "A.y";}
}
class B extends A{
    int i = super.i;
    B(){
      i+=i;
      System.out.println(i);}
    public String x(){return "B.x";}
    public String y(){return "B.y";}
}
class C extends B{
    int i = super.i;
    public String x(){return "C.x";}
}

public class q1{
    public static void main(String[] args) {
        new C();
        A obj1 = new B();
        A obj2 = new C();
        B obj3 = new C();
        System.out.println(obj1.i);
        System.out.println(obj2.i);
        System.out.println(obj3.i);
        System.out.println(obj1.x());
        System.out.println(obj2.x());
        System.out.println(obj3.y());
}}
```

2. What output is produced by the following code? (10points, no partial credits)

```
import java.util.*;
public class q2 {
   public static void main(String[] args) {
     HashSet<Integer> hs = new HashSet<>();
     for (int i=1; i<=4; i++) {
       hs.add(i);
       if (i / 2 == 0) {
        hs.add(i);
        }
     }
     Iterator hi = hs.iterator();
     int sum=0;
     while(hi.hasNext()){
        sum += (int)hi.next();
     }
     System.out.println(sum);
    }}
```

3. What output is produced by the following code? (10points, no partial credits)

```
class Super {
  int a = 1;
  int b = 2;
 public int getSum(int a, int b) {
    this.a+=a;
    this.b+=b;
    return this.a + this.b;}
}

class Sub extends Super {
  int b = 10;
  public int getSum(int a, int b) {
    this.a=a;
    this.b=b;
    return a+b;}
}

public class q3 {
  public static void main (String[] args) {
   Sub obj1 = new Sub();
   Super obj2 = new Sub();
   System.out.println(obj2.getSum(obj1.b, obj2.b));
   System.out.println(obj2.getSum(obj1.a, obj2.a));
   }
}
```

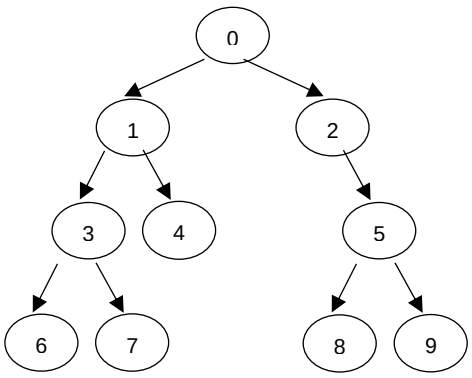
4. Given the input stream 1, 2, 3, 4, 5 Write the sequence of operations (Push(value) for stack and Pop for unstack) which would produce the sequence 2, 3, 5, 4, 1. (5points, no partial credits)

5. How can we retrieve the min element from a min-heap at any time in O(1) time complexity? Explain with an example. (10points)

6. Why or why not LinkedList over ArrayList for Insertion and Deletion? Explain with an example. (10points)

7. Explain with an example how to determine node has right child or not in min heap tree. (10points)

8. Find pre-order, in-order and post-order successor. (4 points/each)



9. Create a tree that has the following traversal. (10points, no partial credits)

```
in-order: a + b * c + d / e
post-order: a b c * + d e / +
```

10. Using big-O notation, how much time does the following each method take? (2points/each)

```
1)    for (int i=0; i<k; i++) {
       }

2)    for(int i= 0; i< 20; i++) {
        for(int j= 1; j< n; j *= 2){
            System.out.println(i);}
      }

3)  for (int i = 0; i < n; i++) {
        if(i < n/2){
            System.out.println("less than half ");
        }
    }

4)  int cur = 1;
    while (cur < n) {
      cur = cur * 2;
    }
```

11. What output is produced by the following code? (10points, no partial credits)

```
import java.util.*;
public class q11 {
 public static void main(String[] args) {
 ArrayList<Integer> al = new ArrayList<Integer>();
 HashSet<Integer> hs = new HashSet<Integer>();
  for (int i= 0; i<5; i++) {
     al.add(i);
     al.add(i/2);
     hs.add(i);
     hs.add(i/2);
     hs.add(i/3);
  }
  System.out.println(al);
  System.out.println(hs);
 }
}
```

12. Draw the tree state of a min-heap after adding these elements. You should draw the tree step by step (NOT a FINAL ONE). (15points)

`2, 5, 1, 4, 6`

13. Draw the above tree state of a min-heap after eliminating <u>one min-value</u>. You should draw the tree step by step (NOT a FINAL ONE). (15points)

14. Modify the following code using java collections (container) so that it produces the same output. You should not use if-statement or switch-statement. (10points)

```
public class q14 {
static int evaluate(char c) {
      if(c == '+') {
            return 1;
      }else if(c == '-') {
            return 1;
      }else if(c == '*') {
            return 2;
      }else if(c == '/') {
            return 2;
      }
   else return 0;
   }

 public static void main(String[] args){
      System.out.println(evaluate('+'));
     System.out.println(evaluate('*'));
 }
}
```

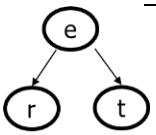
15. What is greatest weakness of the following code? Explain the weakness in detail and modify the code to overcome the weakness. (10points)

```
class minHeap {
   public int howMany;
   private int[] myArray;
   public minHeap() {
      howMany = 45;
      myArray = new int[howMany];
   }
}
public class q16WeakPoint {
   public static void main(String[] args) {
      minHeap mh = new minHeap();
   }
 }
```

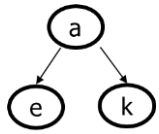
16. Execute the sequence of the following statements. Answer all the four println statements, and leave traces of solving this question. (25points)

```
String s ="I ";
push("u"); push("o"); push("s");
s = s + pop();
push("h");
s = s + pop(); s = s + pop(); s = s + pop();
push("d"); push("l");
s = s + pop(); s = s + pop(); s = s + " ";
System.out.println(s);
```

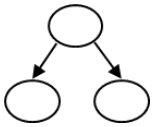
s = s + in-order of the following tree



Convert the following binary tree to binary search tree



<binary tree>



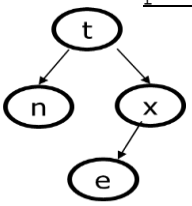
<binary search tree>

s = s + post-order of the above binary search tree

```
s = s + " ";
System.out.println(s);
```

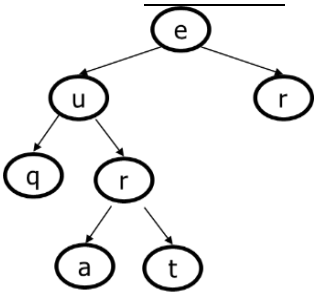
```
Set<String> kim=new HashSet<String>(); Set<String> lee=new HashSet<String>();
kim.add("cs211");  kim.add("cs210");
lee.add("cs211");  lee.add("cs212");
Set<String> kimANDlee = new HashSet<>(kim);
kimANDlee.retainAll(lee);
s = s + kimANDlee;
s = s + " ";
System.out.println(s);
```

s = s + post-order of the following tree




```
s = s + " ";
```

s = s + in-order of the following tree




```
System.out.println(s);
```